AI FOR SCIENCE · AIAS+ STUDENT SHOWCASE

PLANET PIONEERS · AIAS+ 2026 STUDENT SHOWCASE

ExoPlain: An Interactive, Explainable AI Platform for Exoplanet Discovery

An accessible workflow that connects exoplanet measurements with rapid, plain-language AI explanations.

Planet Pioneers · Student research team · NASA Space Apps Challenge 2025

RESEARCH QUESTION

How might an interactive AI tool help students explore exoplanet evidence without confusing an AI-generated explanation with a scientific confirmation?

1 Why this project?

Exoplanet discovery draws on complex signals: a planet passing in front of a star can produce a small, repeating dip in the star's measured brightness. The data are fascinating, but the parameters and classification process can be difficult for new learners to interpret.

Goal: build an approachable learning experience that connects candidate measurements to plain-language explanations and interactive exploration.

The design challenge: make the science easier to explore without presenting an unvalidated AI guess as a discovery.

2 How the prototype works

TWO RELATED PATHS—NOT ONE VALIDATED CLASSIFIER

Explore — Browse example planets in an interactive 3D galaxy map

› **Enter** — Submit Kepler-style transit and stellar measurements

› **Explain** — Gemini API returns a generated classification and feature summary

Upload — Optional FITS light-curve upload in the web prototype

› **Analyze** — Python service is intended to extract signal features

› **Experiment** — Separate Python scripts explore supervised models

The web app's Gemini analysis and the Python TESS model experiments are separate paths in the current codebase.

3 Data & methods

- **Web prototype:** Next.js / TypeScript interface for orbital, transit, planetary, stellar, and observational inputs.
- **AI explanation:** the server sends submitted measurements to Gemini and requests a structured answer with important features.
- **Light-curve experiments:** the Python project includes preprocessing and multiple model experiments (including CNN, LSTM, random-forest, and MLP approaches).
- **Sequence input:** the training scripts use 2,048-point light-curve inputs; one training path uses an 80/20 train-validation split and augmentation.

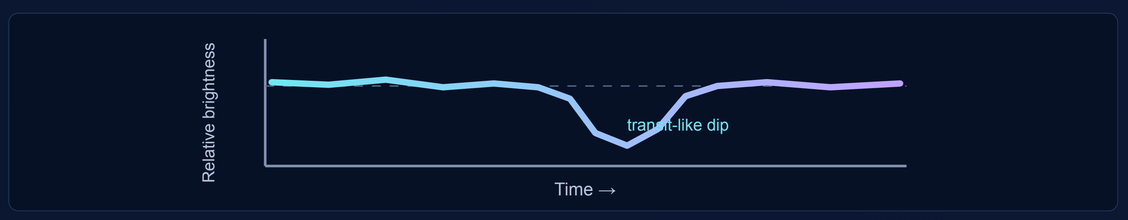
Orbital period · Transit depth · Transit duration · Planet radius · Stellar temperature · Light curve

Data note: project documentation reports 7,638 TESS light curves for a CNN experiment. Dataset provenance, label definitions, and split-level counts should be verified before treating that number as a reproducible study result.

4 What the prototype demonstrates

The codebase brings together a space-themed interface, a 3D example-planet explorer, structured candidate-data entry, a Gemini-powered explanation route, and optional FITS upload UI.

Its central contribution at this stage is **an educational interaction design**: moving from measurements to a readable explanation, while showing where a learner should ask further questions.



Conceptual sketch only; not a measured light curve or model output from this project.

5 Current findings

- **Built:** an interactive web prototype with data-entry, example exploration, and AI-explanation components.
- **Explored:** several Python machine-learning architectures and preprocessing workflows for light-curve classification.
- **Not established:** the available project materials do not provide an independently reproducible, held-out accuracy evaluation for the complete user-facing system.

Interpretation: this is a prototype and model-development project—not evidence that Exo-Plain can confirm planets or outperform expert review.

6 Limitations & next steps

- **Confidence is not calibrated:** the Gemini response is converted to fixed 85% / 15% confidence in the current API code.
- **Fallbacks can be simulated:** FITS analysis may return randomized demo features when the Python service is unavailable.
- **Need a rigorous benchmark:** verify labels and provenance, prevent target leakage with a documented split, then report accuracy, precision, recall, F1, and a confusion matrix.
- **Improve scientific safeguards:** separate candidate triage from confirmation, cite the source data, expose uncertainty, and test explanations with students and astronomy mentors.

TAKE-HOME MESSAGE

AI can make exoplanet data more approachable, while trustworthy science still requires verified labels, reproducible evaluation, calibrated uncertainty, and human expertise.

Project sources: ExoPlain Next.js application and API routes; ExoplanetWebsiteFiles Python training scripts and README; NASA Space Apps Challenge 2025 project page.

NEXT MILESTONE

Audit the dataset, run a reproducible benchmark, and revise the interface to communicate uncertainty clearly.

Project: spaceappschallenge.org/2025/find-a-team/planet-pioneers2/